

Sinjini Dass

ECON238-Homework 4

Github Link: <https://github.com/sdass3/econ238-portfolio>

WEBSITE LINK: <https://claude.ai/artifact/RVJBIFwPS8bL6LzABGz1vq>

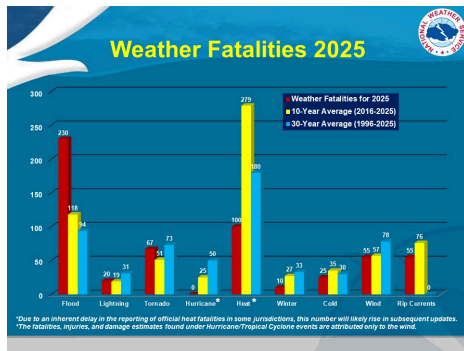
I could not get this onto Github for the life of me.

EETT Brainstorm Log #2:

Similar to last week, we continued discussing data centers, but this conversation focused more on their economic costs. Our group was engaged and discussed whether the benefits of data centers, such as jobs and investment, justify the costs they create. One concern I raised was who should pay for the additional infrastructure they require. After our discussion, I found that in Ohio, SB Energy agreed to invest \$4.2 billion in transmission upgrades for a planned data center, rather than passing those costs onto nearby residents. The project is expected to create over 10,000 construction jobs over four years. This made me think about opportunity cost: while those jobs are valuable, do these jobs justify the costs? It also raises an incentive question: if data centers pay the full marginal cost of the infrastructure they require, would that affect which communities are being impacted (due to relocation).

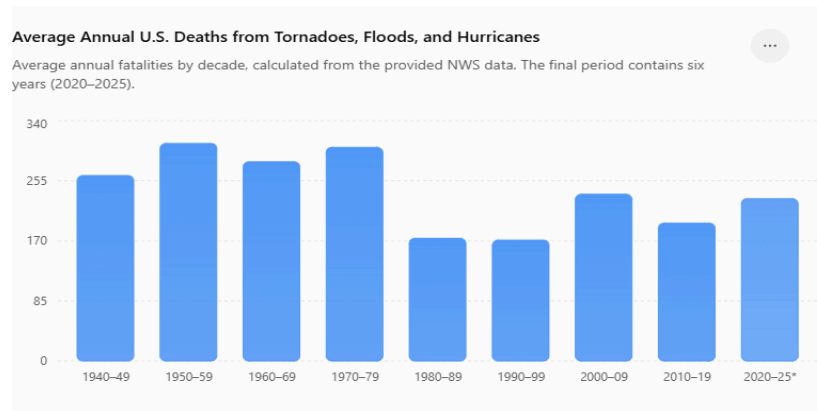
One aspect of data centers that I had not really thought about before was the infrastructure needed to support them. When people discuss data centers, the conversation usually focuses on AI, water usage, or the jobs they create. I had not considered that a data center may require billions of dollars in transmission upgrades simply to supply it with enough electricity. The number is frightening, coming from someone who wasn't necessarily against data centers. The Ohio example made me think differently about the issue because the question becomes not only how much electricity a data center uses, but who pays for the infrastructure needed to provide it. If the company pays for these upgrades, its private costs are higher, but if other electricity customers or taxpayers pay for them, some of the costs are shifted onto people who may not directly benefit from the project. It feels like unnecessary money.

Research:



A.

- a. Link to source: https://www.weather.gov/media/hazstat/81year_2025.pdf



b.

B. Avg number of fatalities

- 2025-1940, inclusive of both = 86 years
- To find the avg number of fatalities, I take the avg number of fatalities per year, multiply each by 10 (except for 2020s) and sum them up. Then divide by 86
- The 2020s isn't done yet so it's multiplied by 6 instead of 10
- The answer is roughly 232 deaths

C. Based on data

- Solely based on the data, it doesn't look like there's any specific trend upward. There is a decline from about 300 deaths per year earlier on, that drifts more towards the 230s in the 80s and 90s onward. Adding heat and cold related deaths could change things, but I think there's a problem more so with heat deaths that they are counted as other things such as heart failure or maybe cardiac arrest when it's cold and someone is shoveling snow. Perhaps that's why they are not included.

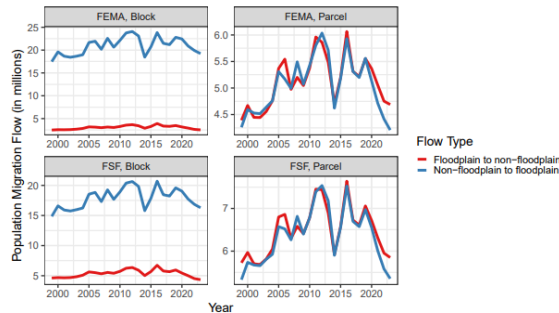
D. Population

- Using this source:
<https://www2.census.gov/programs-surveys/popest/tables/1900-1980/national/totals/popclockest.txt>
- Annual average deaths for weather event/average population x 100,000

- c. $1940: 300/140,000,000 (100,000)=0.21$ deaths per 100 people

E. Hazard: Flooding

Figure 3: Migration Flows To and From Floodplains, Block vs. Parcel Exposure



a.

b. Link:

<https://www2.census.gov/library/working-papers/2026/adrm/ces/CES-WP-26-37.pdf>

- c. The above figure looks at the migration flows between floodplains and non-floodplain in the US between 2000 to 2020, so a bit outdated. Two measures of exposure are compared, if a person lives in a floodplain and the census block level and whether their parcel is exposed. If one looks at it broadly, migration towards floodplains seems like it's increased over time, especially in the first figure. On the right, for the parcel level, the gap is significantly smaller. Looking at this data and modeling, it seems that Americans are moving to floodplain locations though it does not automatically mean they're exposing themselves to flooding. This is key because it's easy to assume that moving to a floodplain area does not mean the properties are at high risk, though it might seem like a logical conclusion to make. Weather related fatalities have declined as shown in the earlier analysis, likely not because these events have become more rare but because Americans have developed better protection from hazards like flooding. With better technology, it is easier now more than ever to save oneself from these disasters, so more people can move to "more prone" areas.

F. Structures

- Assume avg lifetime of 35 years
- $\$600 \text{ Billion} \times 0.15 \text{ (relocation cost assumption)} \times 1/35 = \$2,571,428.57$ per year
- This should still be a very small portion of GDP $2.57 \text{ Billion} / 30,000 \text{ Billion} =$ approx 0.01% of GDP

G. Summary

- The IPCC's report reports that increasing disaster losses are connected with larger exposure and vulnerability because more people and their assets are in hazard prone areas. Like in my flood analysis, people are not just avoiding moving to hazard prone areas, they are doing so but with precautions. With larger groups in these areas, larger disasters can happen.

- H. Heat related deaths have more variability. Heat mortality doesn't just depend on the temperature itself, but also weather conditions, age, health etc. This is key for the coming years because it's important to know that there are other omitted variables like the ones mentioned above. Higher temperatures does not necessarily mean more heat related deaths with the continued use of related technology.

Research and Reflection:

Diet Juice

- A. I don't really know much about these costs so I had to google
- Electricity used per charge: $15\text{Wh}/0.90$ (90% efficiency) $= 0.0167(356 \text{ for year}) = 6.08 \times 0.1966 = \$1.20/\text{year}$
 - This number is a lot smaller than I anticipated, I feel kind of good about it actually.
- B. $2.31 \rightarrow \text{Co}_2 \text{ per kWh from internet}$
- $6.08 \times 2.31 = 14.1$
 - Converting to metric tons must divide by 2,204.6
 - $14.1/2,204.6 = 0.00637$
 - Now for carbon damage about \$200 per metric ton of damage
 - $200 \times 0.00637 = \$1.27 \rightarrow \text{cost to charge phone with coal}$
- C. If all 8 billion people did it
- $1.27(8,000,000) = 10,160,000$
 - Cost would be at least 10 billion dollars

Cool down:

I opted to have a recorded discussion, see my Github.

#4. Our discussion touched on a lot of the points and facts that ended up being true, but we weren't sure why. For the question of why Europe experiences more heat-related deaths, my gut instinct was that Europeans simply don't use air conditioning as much. It's a weak argument on its own and without the internet, that was all I could come up with, and it doesn't explain *why* AC adoption differs or whether it's even the main driver. Burke actually complicates or rejects this more than it confirms it. The paper finds that cold kills more than heat everywhere in their sample, and that higher income isn't uniformly protective against temperature which undercuts the simple "just install AC" fix. So my instinct wasn't wrong that AC infrastructure matters, but Burke suggests that it is less about one fix and more about how exposure, age etc interact. The 5–12% figure is also interesting against our earlier storm-death data. That's a large share of total deaths compared to the lower hundreds of deaths from tornadoes, floods, and hurricanes

combined in an average year. The difference is NWS storm counts capture deaths directly caused by a specific acute event, while Burke's number captures conditions and after effects caused by these events. One intervention worth considering is government-subsidized air conditioning or cooling access for low-income households. Economically, the cost is the subsidy itself plus electricity and maintenance, compared against the next-best use of that public money and Burke's finding that many interventions lack empirical support means we can't just assume subsidized AC delivers proportional mortality reduction; someone would need to actually measure it. For the reason I am unsure of its effectiveness. But, for my hold on, if temperature-related deaths aren't falling over time like Burke found, is that because adaptation isn't working, or because more people who aren't used to the heat are now exposed to it?